

Table 1: State-of-the-art Methods for Semi-Supervised Clustering

	Initialization		Assignment		Solution Mutation and Crossover		
	Greedy Method	ILP	Greedy Method	ILP	Centroid Perturbation	Centroid Reposition	Centroid Crossover
KSKM	DSATUR-based	No	Kempe Swaps	No	Yes	Yes	No
KSKM_E	DSATUR-based	Yes	Kempe Swaps	Yes	Yes	Yes	No
PCCC	No	Yes	No	Yes	No	Yes	No
BLPKMCC	No	Yes	No	Yes	No	No	No
S_MDEClust	Yes	Yes	By order of data	Yes	No	Optional	Yes
COPKM	Yes	No	By order of data	No	No	No	No
COPKM_P	DSATUR-based	No	By order of DSATUR	No	No	No	No